SEEDLINE · REGULATORY AND STANDARDS REGISTER

SeedLine regulatory and standards register

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DOCUMENT CONTROL

Document ID	SDL-REG-001	Version	0.1
Status	Draft	Date	2026-10-24
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/seedline		
Source	docs/07-regulatory.md @ 0a57933		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 pipeline, kit 1.9.1 upgrade), completed for the current design (option A) at stage P3

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No product certification is known to apply to a hand-pushed seeder: the EU machinery rules start from a drive system "other than directly applied human or animal effort" (G1), and ISO 4254-1 covers self-propelled, mounted, semi-mounted and trailed agricultural machines (G2). The test method for single-seed drills (G3) is the reference for the placement requirements, the guarding reference (G4) and the manual-handling references (G5, G6) are used as good practice, and treated seed is controlled by its label and national pesticide rules (G7). National rules in the partner's country are not yet known (G8). Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Nothing here needs certification before TRL 4. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Directive 2006/42/EC (Machinery Directive), Article 2(a); Regulation (EU) 2023/1230 (Machinery Regulation)	European Union	Safety of machinery placed on the market; "machinery" is an assembly "fitted with or intended to be fitted with a drive system other than directly applied human or animal effort"	SeedLine is pushed by hand and has no other drive system, so it appears to fall outside the definition; to confirm for the Regulation's text and for any later powered variant	Design lead	Does not apply (to confirm with a qualified reviewer)
G2	ISO 4254-1:2013, Agricultural machinery, Safety, Part 1: General requirements	International	Safety requirements for self-propelled ride-on, mounted, semi-mounted and trailed agricultural machines	A pedestrian-pushed seeder is not in the stated scope; used only as a reference for guarding and marking practice	Design lead	Does not apply (reference only, to assess)
G3	ISO 7256-1:1984, Sowing equipment, Test methods, Part 1: Single seed drills (precision drills)	International	Test methods for seed spacing (misses, multiples, uniformity)	PR-001 and PR-002 are written in its terms; the TRL 4 bench test and the field counts follow it	Test lead (TRL 4)	Applies (method; tests on hold at the TRL 3 cap)
G4	ISO 13857:2019, Safety of machinery, Safety distances to prevent hazard zones being reached by upper and lower limbs	International	Guard opening sizes and reach distances	The chain guard (PR-014); the guard openings are not yet checked against its tables	Design lead	To assess (good practice; not mandatory for an unpowered seeder)
G5	ISO 11228-2, Ergonomics, Manual handling, Part 2: Pushing and pulling	International	Recommended force limits for pushing and pulling	The design-case push of 132 N horizontal, 206 N along the handle (PR-010) and whether one operator sustains it over a planting day (SDL-STM-001 row S9)	Design lead	To assess
G6	ISO 11228-1, Ergonomics, Manual handling, Part 1: Lifting and carrying	International	Mass limits for manual lifting	14.8 kg with the marker kit, lifted and turned by one person at row ends (PR-011)	Design lead	To assess
G7	Pesticide product labels and national pesticide regulations for treated seed	Country of use	Personal protective equipment, handling and disposal of seed dressed with fungicide or insecticide	Filling, emptying and cleaning the hopper and housing (UR-017, PRC Safety)	Co-design partner; user	Applies (follow the label); national rules to assess
G8	National agricultural machinery testing or			Unknown until the partner and country are		To assess

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	standards schemes in the partner's country (for example national standards bodies in Eastern and Southern Africa)	Country of deployment	Testing or certification of hand planters, where a scheme exists	fixed (first candidate: CIMMYT small-scale mechanization work in Eastern and Southern Africa, SDL-DEC-001)	Co-design partner	
G9	CERN-OHL-S-2.0 and MIT licences; REUSE specification	Open hardware practice	Licensing of the design files and software	Every file in the repository (`REUSE.toml`)	Design lead	Met by design

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- Directive 2006/42/EC of the European Parliament and of the Council (Machinery Directive), Article 2, https://www.legislation.gov.uk/eudr/2006/42/article/2/data.htm, accessed 2026-10-24.
- Regulation (EU) 2023/1230 (Machinery Regulation), https://eur-lex.europa.eu/eli/reg/2023/1230, accessed 2026-10-24 (definition not re-read for this register; G1 to confirm).
- ISO 4254-1:2013 scope as published by the Swedish Institute for Standards, https://www.sis.se/en/produkter/agriculture/agricultural-machines-implements-and-equipment/general/iso425412013/, accessed 2026-10-24.
- ISO 7256-1:1984, title and status, https://www.iso.org/standard/13910.html, accessed 2026-10-24.
- ISO 13857:2019, title as listed by Genorma, https://genorma.com/en/standards/iso-13857-2019, accessed 2026-10-24.
- International Committee of the Red Cross, *Warning: Treated Seeds*, https://www.icrc.org/en/publication/4688-warning-treated-seeds, accessed 2026-10-24.
- Standards G5 and G6 are cited by number and title only; their texts were not consulted for this register, and each applicability is to be confirmed by the role named.